

Tyrel Boese

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EDUCATION

Colorado Mesa University B.Sc. in Physics, Minor in Math

Expected May 2026

RESEARCH EXPERIENCE

Research Internship, Center For Quantum Networks, University of Arizona, 2025

Project: Design of integrated photonic circuits using heterogenous materials.

Senior Research, Colorado Mesa University, 2024-2025

Project 1: Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

Project 2: Multiscale Modeling of Ultrafast Laser-Induced Melting and Metastability in Gold

Project 3: Applications of Quantum Machine Learning and the Variation-Quantum Eigensolver in determining adiabatic and non-adiabatic exchange correlation functionals

POSTER PRESENTATIONS

- **Rocky Mountain Advanced Computing Consortium High Performance Computing Symposium Poster Competition**

Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

Authors: Tyrel Boese, Ian Anderson, James Goff, Jarrod Schiffbauer

Date: Spring 2025

1st Place Award

- **Colorado Mesa University Student Showcase**

Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

Authors: Tyrel Boese, Ian Anderson, James Goff, Jarrod Schiffbauer

Date: Spring 2025

- **Colorado Space Grant Consortium**

Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold

Authors: Tyrel Boese, Ian Anderson, James Goff, Jarrod Schiffbauer

Date: Spring 2025

- **CQN REU Virtual Poster Presentation**

Designing a Waveguide for Integrated Photonics Applications

Authors: Tyrel Boese, Neshat Sadafi, Matt Eichenfield

Date: Summer 2025

SEMINAR PRESENTATIONS

- *Designing a Waveguide for Integrated Photonics Applications*, Colorado Mesa University Physics Seminar, Fall 2025
- *Progress Towards a Quantum-Accurate Classical SNAP Machine Learning Interaction Potential for Gold*, Colorado Mesa University Physics Seminar, Spring 2025

HONORS AND AWARDS

- 1st Place, Rocky Mountain Advanced Computing Consortium (RMACC) High Performance Computing (HPC) Poster Competition (2025).
- NASA Bridge Program Seed Fund (BPSF) Scholar

LEADERSHIP

- Engineering Workforce Development Officer, Center For Quantum Networks (CQN) Student Post-Doc Leadership Council (SPLC)
- Vice President, Colorado Mesa University Society of Physics Students

AFFILIATIONS

- American Physical Society (APS) Member
- Society of Physics Students (SPS) Member

SKILLS

- Analytical and Mathematical Problem Solving
- Programming: Python, Fortran, C++, C#, C, MATLAB, Julia, R
- Tools: LAMMPS, Quantum Espresso, FitSNAP, COMSOL, Git, DFT-Julia

WORK EXPERIENCE

SMB Account Executive, Spot-On Technologies, 2023

Multimedia Account Executive, Gray Media, 2022

SMB Account Executive, Spectrum Business, 2020

ACADEMIC INTERESTS

- Quantum Computing - Quantum materials, Applications in Quantum Chemistry, Variational Quantum Eigensolvers
- Numerical Modeling - Molecular Dynamics, Machine Learning Force Field Interatomic Potentials, Multiscale Simulations
- Procedural Generation - Noise Functions, Simplex Noise, Perlin Noise.

REFERENCES

Available upon request.